

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: COMPOSITION CONTROL, BARRIER STRUCTURES, AND ELASTOMERIC APPLICATIONS

1. ABSTRACT

This report surveys a five-patent landscape relevant to low acrylonitrile nitrile-butadiene rubber (NBR) and adjacent elastomeric technologies. The target is defined as NBR with reduced acrylonitrile content or composition control over acrylonitrile/butadiene feed ratio, with emphasis on industrially relevant synthesis, emulsion or solution copolymerization, and downstream performance in hoses, seals, and high-demand elastomeric components. Across the manifest, only one patent directly addresses NBR as a hose material (EP2438340B1), while the others provide adjacent formulation or application context rather than direct NBR polymerization routes.

The strongest direct relevance comes from EP2438340B1, which identifies NBR among preferred non-fluorinated rubber inner-tube materials for flexible fuel hose and pairs it with a branched PA6 barrier layer and impact modifier to achieve low permeation and improved flexibility.

US10443339B2 is also relevant at the application level, describing elastomeric components for bore tools and packers in harsh environments and explicitly naming NBR among candidate elastomers, though it focuses on nanoplatelet reinforcement rather than NBR synthesis.

US20200239671A1 is a wear-resistant rubber composition patent centered on natural rubber, polybutadiene, and hydroxy-terminated polybutadiene; it is only indirectly relevant as a formulation/process benchmark for durable elastomer systems. US7871546B2 concerns vegetable-oil dielectric fluids and is not chemically related to NBR synthesis, but it illustrates the broader patent environment for elastomer-adjacent industrial materials. US20080108752A1 discloses controlled free-radical block copolymerization of styrene and maleic anhydride using TEMPO/BPO chemistry; it is not an NBR patent, but it is useful as a process reference for controlled copolymerization, monomer-feed management, and molecular-weight control.

Overall, the evidence suggests that low-acrylonitrile NBR is most often treated as a materials-selection problem in end-use patents rather than a dedicated synthesis patent in this set. The most transferable process themes are composition control, barrier-layer integration, and reinforcement/compatibilization strategies rather than explicit low-ACN polymerization recipes.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20200239671A1
- Patent Title: Wear-Resistant Rubber Compositions, Systems, And Methods
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses wear-resistant rubber compositions built from natural rubber, polybutadiene, hydroxy-terminated polybutadiene, sulfur, accelerants, silica, and optional additives. The compositions are compounded and cured to achieve high crosslink density and improved wear performance in mining-related rubber parts such as lifter bars and shock absorbers. - Natural rubber: about 20 wt. % to about 70 wt. % — claim 51 - Hydroxy-terminated polybutadiene: about 1 wt. % to about 10 wt. % — claim 51 - Hydroxy-terminated polybutadiene: about 2 wt. % — synthesis passage - Polybutadiene: about 2 wt. % to about 20 wt. % — synthesis passage - Effective cross-linking density: at least about 30×10^{-5} moles/cm³ — claim 48 - Effective cross-linking density: at least about 40×10^{-5} moles/cm³ — synthesis passage - Effective cross-linking density: at least about 60×10^{-5} moles/cm³ — synthesis passage - Wear period: about 1 week, about 3 weeks, about 7 weeks, about 14 weeks, about 21 weeks, and about 24 weeks — synthesis passage - Wear: about 10% to about 40% — synthesis passage - Wear: at most about 30% over about 24 weeks — synthesis passage - Lifter bar wear test duration: four weeks — Example 2 - Rheometer MDR: 4 minutes @ 350° F., 1° Arc — Example 1 - Samples cured: 30 minutes at 280° F. — Example 1 - Soxhlet swelling: 24 hours at 120° F. using tetrahydrofuran — Example 4 - Crosshead displacement rate: 0.02 inch per minute — Example 4

Relevant Experimental Evidence - Example 1: Nine rubber compositions with 0 to 8 phr hydroxy-terminated polybutadiene were compared for rheology and tensile/tear properties. - Example 2: Lifter bars containing hydroxy-terminated polybutadiene showed less wear over four weeks than compositions without it. - Example 3: A broad formulation matrix (A–P) showed how varying filler, curative, and additive levels changes wear-resistant rubber composition design. - Example 4: A crosslink-density measurement method was demonstrated using swelling in tetrahydrofuran and tensile/compression testing.

Technical Relevance This patent is not an NBR synthesis disclosure, but it is relevant as a benchmark for industrial elastomer compounding and cure design. Its explicit use of hydroxy-terminated polybutadiene, sulfur cure, and crosslink-density control provides formulation context for durable rubber systems that may be compared against low-ACN NBR applications.

Patent 2

- Patent Number: US20080108752A1
- Patent Title: Block copolymers containing functional groups
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2008
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses controlled free-radical copolymerization of styrene and maleic anhydride using a stable free radical (TEMPO derivatives) and benzoyl peroxide initiator. The process is run in bulk or solution and uses the stable-free-radical/initiator ratio and monomer/solvent ratio to control microstructure, molecular weight, and polydispersity. - Reaction temperature: 120° C. — Example I - Reaction vessel: 20 ml quiescent vials — Example I - Maleic anhydride: 5 wt. % — Example I - Maleic anhydride: 10 wt. % — Example I - Maleic anhydride: 15 wt. % — Example I - Benzoyl peroxide (BPO): 0.033 M — Example I - HO-TEMPO/BPO molar ratio: 1.3 — Example I - Reaction time: 0.333 hr — Example I - Reaction time: 1 hr — Example I - Reaction time: 3 hr — Example I - Reaction time: 5 hr — Example I - Reaction temperature: 120° C. and 130° C. — Example II - Reaction volume: 3 ml — Example II - [oxo-TEMPO]/[BPO]: 1.3 — Example II - Initial [BPO]: 0.003 M — Example II - BPO: 0.0065 M — Example V - Reaction temperature: 120° C. — Example V - Monomer/solvent ratio: 50-50 — Example V, Tables VIII and IX - Monomer/solvent ratio: 75-25 — Example V, Tables VIII and IX - [NOx]/[IO]: 1.3 — Example V, Tables VIII and IX - [NOx]/[IO]: 1.9 — Example V, Tables VIII and IX - Maleic anhydride wt. %: 0.5 wt. % — Example V, Tables VIII and IX - Maleic anhydride wt. %: 2 wt. % — Example V, Tables VIII and IX - Maleic anhydride wt. %: 5 wt. % — Example V, Tables VIII and IX - Reaction time: 1 hr — Example V, Tables VIII and IX - Reaction time: 10 hr — Example V, Tables VIII and IX - Reaction time: 17 hr — Example V, Tables VIII and IX

Relevant Experimental Evidence - Example I: Bulk copolymerization at 120° C. showed increasing conversion and molecular weight with time and MA content. - Example II: Parallel reactor experiments at 120° C. and 130° C. demonstrated controlled molecular-weight growth over time. - Example V: Solution polymerization in toluene or xylene showed the effect of monomer/solvent ratio and nitroxide/initiator ratio on conversion and polydispersity.

Technical Relevance Although not an NBR patent, it is highly relevant as a controlled copolymerization reference for composition and molecular-weight control. The patent demonstrates how feed ratio, stable free radical chemistry, and reaction conditions can be tuned to manage copolymer microstructure, a transferable concept for low-acrylonitrile NBR feed-control strategies.

Patent 3

- Patent Number: US10443339B2
- Patent Title: Elastomeric materials
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses elastomeric components for bore tools and packer assemblies in harsh downhole environments. The elastomeric matrix is reinforced with carbon-based nanoplatelets and may be combined with various elastomers, including NBR, to improve performance under high temperature, pressure, and chemically aggressive conditions. - Carbon-based nanoplatelets: less than approximately 10 percent by weight — claim 1 - Temperature: about 160 degrees C. — background passage for HNBR property loss - Temperature: about 220 degrees C. — background passage for AFLAS property loss - Pressure: up to about 138 MPa — background passage for HPHT environment - Temperature: up to about 205 degrees C. — background passage for HPHT environment - Pressure: up to about 241 MPa — background passage for ultra-HPHT environment - Temperature: up to about 260 degrees C. — background passage for ultra-HPHT environment - Temperature: about 3 MPa — background passage for ambient rubber modulus

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it explicitly identifies NBR as a candidate elastomer for high-performance downhole seals and packers. While it does not teach NBR synthesis, it shows the application space where low-ACN NBR composition choices may matter for chemical resistance and mechanical retention.

Patent 4

- Patent Number: US7871546B2
- Patent Title: Vegetable oil based dielectric coolant
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses dielectric fluid compositions based on vegetable oils or vegetable-oil blends for electrical equipment, with antioxidant compounds and limited petroleum-derived mineral oil content. It is a fluid-formulation

patent rather than a rubber synthesis patent, and it does not disclose NBR preparation. - Vegetable oil viscosity: between 2 and 15 cSt at 100° C. — claim 1 - Vegetable oil viscosity: less than 110 cSt at 40° C. — claim 1 - Petroleum-derived mineral oil: less than 50% by weight of the dielectric fluid — claim 1 - Petroleum-derived mineral oil: less than 30% by weight of the dielectric fluid — claim 4 - Petroleum-derived mineral oil: less than 20% by weight of the dielectric fluid — claim 5 - Open-cup fire point: equal to or greater than 300° C. — synthesis passage - Water saturation point of mineral oils: approximately 65 ppm at room temperature — synthesis passage - Water saturation point of mineral oils: over 500 ppm at normal operating temperatures (approx. 100° C.) — synthesis passage - Moisture removal target: below about 35 ppm — synthesis passage - Moisture removal prior to filling: between about 15 and 25 percent saturation at room temperature (10-15 ppm) — synthesis passage

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it concerns dielectric fluids, not elastomer synthesis. It is included in the manifest as adjacent industrial materials context, but it does not contribute actionable NBR polymerization or composition-control data.

Patent 5

- Patent Number: EP2438340B1
- Patent Title: Low-permeation flexible fuel hose
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a flexible fuel hose with an elastomer inner tube, an elastomer outer cover, and an intermediate barrier layer comprising branched polyamide 6 with impact modifier. The description explicitly identifies NBR as a preferred non-fluorinated rubber for the inner tube and discusses multilayer hose construction to reduce fuel permeation and improve flexibility. - Carbon-based nanoplatelets: less than approximately 10 percent by weight — claim 1 - Flexural modulus of PA 6: from 1 to 2 GPa — claim 5 - Tensile elongation at break of PA 6: 100% or more — claim 6 - Radial thickness of barrier layer: 0.025 mm to 0.76 mm — claim 7 - Permeability to CM15 or CE10 Fuel: less than 2 g/m²/day at 40°C — claim 10 - Permeability to CM15 or CE10 Fuel: less than 40 g/m²/day at 60°C — claim 10 - Permeability to CM15 or CE10 Fuel: less than 20 g/m²/day — claim 10

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is directly relevant to low-acrylonitrile NBR because it explicitly names NBR as a suitable non-fluorinated rubber for the hose inner tube. It shows how NBR is used in a multilayer architecture with PA6 barrier design to meet low-permeation fuel-hose requirements, which is a key downstream application for low-ACN NBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Only EP2438340B1 directly uses NBR in a target application; the other patents are adjacent materials or process references rather than low-ACN NBR synthesis disclosures.
- US20080108752A1 provides the clearest composition-control analogy: monomer ratio, stable-free-radical/initiator ratio, temperature, and time are explicitly tuned to control copolymer microstructure and molecular weight.
- US20200239671A1 and EP2438340B1 are formulation/application patents, emphasizing compounded elastomer performance rather than monomer feed control; both rely on multi-component rubber systems and property targets.
- US10443339B2 focuses on harsh-environment elastomer service and explicitly lists NBR as a candidate elastomer, but the disclosed innovation is nanoplatelet reinforcement, not acrylonitrile-content adjustment.
- US7871546B2 is outside the NBR chemistry space and contributes only general industrial materials context; it does not disclose polymerization, emulsifier, initiator, or chain-transfer conditions relevant to NBR synthesis.

4. CONCLUSION

The manifest contains only one patent that directly names NBR in a relevant product architecture, EP2438340B1, which supports NBR as a non-fluorinated inner-tube elastomer in low-permeation fuel hose. The remaining patents are useful mainly as adjacent references for elastomer compounding, controlled copolymerization, and harsh-environment service requirements. For low-acrylonitrile NBR synthesis specifically, the evidence here does not provide a dedicated polymerization recipe; the most transferable process lesson is from controlled free-radical copolymerization in US20080108752A1, where monomer ratio, stable-free-radical/initiator ratio, temperature, and reaction time govern composition and molecular weight. Based on the evidence, the most defensible synthesis guidance is to prioritize feed-ratio control and tight process control during copolymerization, then validate the resulting low-ACN NBR in multilayer or reinforced formulations for fuel, seal, and downhole applications.

5. REFERENCES

- US20200239671A1 | Wear-Resistant Rubber Compositions, Systems, And Methods | Not disclosed | US | 2020 | <https://patents.google.com/patent/US20200239671A1/en>
- US20080108752A1 | Block copolymers containing functional groups | Not disclosed | US | 2008 | <https://patents.google.com/patent/US20080108752A1/en>
- US10443339B2 | Elastomeric materials | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10443339B2/en>
- US7871546B2 | Vegetable oil based dielectric coolant | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7871546B2/en>
- EP2438340B1 | Low-permeation flexible fuel hose | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2438340B1/en>